Code screenshots:



```java
package textAnalysis;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Scanner;

public class textTool {

    static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args) {
        System.out.println("Enter your text:"); //Enter body of text
        String paragraphBody = scanner.nextLine().toLowerCase();


        int characterCount = paragraphBody.trim().length(); //Character count excluding whitespace
        System.out.println("Character count: " + characterCount);


        int wordCount = paragraphBody.split("\\s").length; //Word count using the split() method
        System.out.println("Word count: " + wordCount);


        System.out.println("Most common character: " + getMostCommonCharacter(paragraphBody));


        System.out.println("\nEnter a character to check its frequency in the text:");
        char character = scanner.nextLine().toLowerCase().trim().charAt(0);
        int frequency = 0;
        for(int i = 0; i < paragraphBody.length(); i++) {
            if(character == paragraphBody.charAt(i)) {
                frequency++;
            }
        }
        System.out.println("Number of character occurrences: " + frequency);


        System.out.println("\nEnter a word to check its frequency in the text:");
        String word = scanner.nextLine().toLowerCase().trim();

        ArrayList<Words> words = new ArrayList<>(); //This ArrayList will be used to check the occurrences of a specified word and store each unique word

        String[] wordArray = paragraphBody.split(" ", 0);
        int i = 0;
        int indexArray = -1;
        for(String str : wordArray) {
            if(i == 0) {
                Words uniqueWord = new Words(str);
                words.add(uniqueWord);
                i++;
                continue;
            }
            for(int j = 0; j <= words.size(); j++) {
                if(j == words.size()) {
                    Words uniqueWord = new Words(str);
                    words.add(uniqueWord);

                    if(str.compareTo(word) == 0) {
                        indexArray = words.size() - 1;
                    }
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                }
                if(str.compareTo(words.get(j).getWord()) == 0) {
                    words.get(j).incFrequency();
                    break;
                }
            }
        }
        if(indexArray == -1) { //Error handling for invalid input
            System.out.println("Invalid input.");
        }
        else {
            System.out.println("Number of word occurences: " + words.get(indexArray).getFrequency());
        }

        System.out.println("Number of unique words: " + words.size()); //Uses the ArrayList of unique words to display the amount
    }


    static char getMostCommonCharacter(String str) { //Method for finding the character that appears the most frequently
        HashMap<Character, Integer> map = new HashMap<>();

        int n = str.length();
        char answer = 0;
        int count = 0;
        for(int i = 0; i < n; i++) {
            char c = str.charAt(i);
            map.put(c, map.getOrDefault(c, 0) + 1);

            if(count < map.get(c)) {
                answer = c;
                count = map.get(c);
            }
        }
        return answer;
    }
}


class Words { //Words class for the ArrayList in the main method
    private String word;
    private int frequency;

    public Words(String word) { //Constructor
        this.word = word;
        this.frequency = 0;
    }

    void incFrequency(){ //Increments the count of how many times a word appears
        frequency++;
    }

    String getWord() { //Return methods
        return word;
    }

    int getFrequency() {
        return frequency;
    }
```

Code text:

```java
package textAnalysis;

import java.util.ArrayList;

import java.util.HashMap;

import java.util.Scanner;

public class textTool {


	static Scanner scanner = new Scanner(System.in);


	public static void main(String[] args) {

		System.out.println("Enter your text:"); //Enter body of text

		String paragraphBody = scanner.nextLine().toLowerCase();


		int characterCount = paragraphBody.trim().length(); //Character count excluding whitespace

		System.out.println("Character count: " + characterCount);



		int wordCount = paragraphBody.split("\\s").length; //Word count using the split() method

		System.out.println("Word count: " + wordCount);



		System.out.println("Most common character: " + getMostCommonCharacter(paragraphBody));



		System.out.println("\nEnter a character to check its frequency in the text:");

		char character = scanner.nextLine().toLowerCase().trim().charAt(0);
```

```
            int frequency = 0;
            for(int i = 0; i < paragraphBody.length(); i++) {
           if(character == paragraphBody.charAt(i)) {
               frequency++;
           }
       }
            System.out.println("Number of character occurrences: " +
frequency);



            System.out.println("\nEnter a word to check its frequency in the
text:");
            String word = scanner.nextLine().toLowerCase().trim();

            ArrayList<Words> words = new ArrayList<>(); //This ArrayList will
be used to check the occurrences of a specified word and store each unique word

            String[] wordArray = paragraphBody.split(" ", 0);
            int i = 0;
            int indexArray = -1;
            for(String str : wordArray) {
                  if(i == 0) {
                        Words uniqueWord = new Words(str);
                        words.add(uniqueWord);
                        i++;
                        continue;
                  }
                  for(int j = 0; j <= words.size(); j++) {
                        if(j == words.size()) {
```

```
                                Words uniqueWord = new Words(str);
                                words.add(uniqueWord);

                                if(str.compareTo(word) == 0) {
                                      indexArray = words.size() - 1;
                                }
                          }
                          if(str.compareTo(words.get(j).getWord()) == 0) {
                                words.get(j).incFrequency();
                                break;
                          }
                    }
              }
              if(indexArray == -1) { //Error handling for invalid input
                    System.out.println("Invalid input.");
              }
              else {
                    System.out.println("Number of word occurences: " +
words.get(indexArray).getFrequency());
              }

              System.out.println("Number of unique words: " + words.size());
//Uses the ArrayList of unique words to display the amount
        }


        static char getMostCommonCharacter(String str) { //Method for finding the
character that appears the most frequently
              HashMap<Character, Integer> map = new HashMap<>();
```

```
      int n = str.length();
       char answer = 0;
       int count = 0;
       for(int i = 0; i < n; i++) {
           char c = str.charAt(i);
           map.put(c, map.getOrDefault(c, 0) + 1);

           if(count < map.get(c)) {
              answer = c;
              count = map.get(c);
           }
       }
       return answer;
   }
}
class Words { //Words class for the ArrayList in the main method
      private String word;
      private int frequency;

      public Words(String word) { //Constructor
            this.word = word;
            this.frequency = 0;
      }

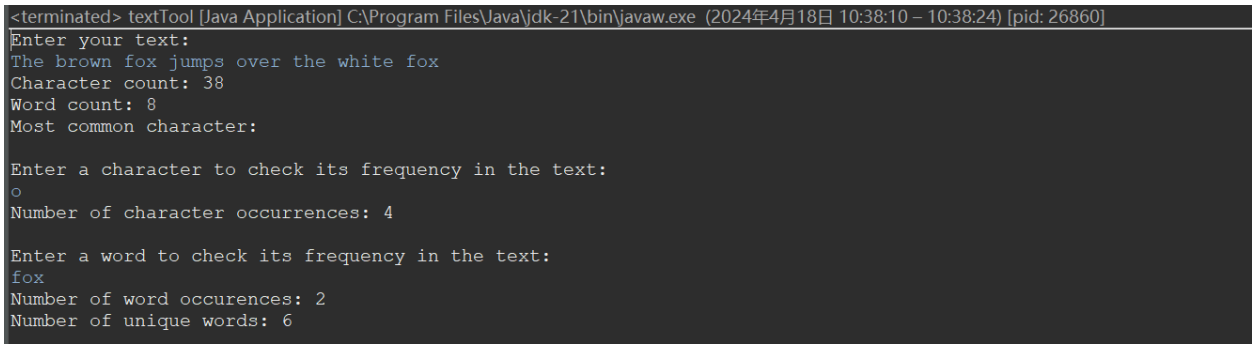
      void incFrequency(){ //Increments the count of how many times a word
appears
            frequency++;
      }
```

```
	String getWord() { //Return methods
		return word;
	}

	int getFrequency() {
		return frequency;
	}

}
```

Output screenshot:



Output text:

```
<terminated> textTool [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (2024年4月18日 10:38:10 – 10:38:24) [pid: 26860]
Enter your text:

The brown fox jumps over the white fox

Character count: 38

Word count: 8

Most common character:
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```
Enter a character to check its frequency in the text:

o

Number of character occurrences: 4

Enter a word to check its frequency in the text:

fox

Number of word occurences: 2

Number of unique words: 6
```